

Auth Service Architecture Guide

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The auth-service is the identity and security foundation of ConnectSphere. It creates accounts, authenticates users, issues JWT access and refresh tokens, supports profile editing, and exposes user-search and token-validation endpoints for the rest of the platform.

Internally it combines Spring Security, JPA entities, repositories, service-layer rules, JWT creation and rotation, and optional Google/GitHub OAuth login.

Architecture Snapshot

Module	auth-service
Type	Spring Boot microservice
Port / Entry	8081
Data Ownership	connectsphere_auth (MySQL) / authdb (H2 tests)
Protocols / Main Tech	REST, JWT, OAuth2, Spring Security
Verification Command	mvn test
Key Endpoints or Routes	/api/v1/auth/register /api/v1/auth/login /api/v1/auth/refresh /api/v1/auth/profile /api/v1/auth/search /oauth2/authorization/google /oauth2/authorization/github

Responsibilities

- User registration and login
- JWT access-token and refresh-token lifecycle
- Profile and password management
- OAuth account onboarding

- User lookup and downstream token validation

Integrations

- Used by search-service for user search
- Supports frontend login/profile screens
- Designed to sit behind the gateway for versioned API access

Internal Working

- Register flow: controller validates the request, service creates the user, password is bcrypt-hashed, token service issues access and refresh tokens, and the API returns the full AuthResponse.
- Login flow: AuthenticationManager verifies credentials, JwtTokenService creates a fresh token pair, and the response contains expiry timestamps plus the user profile.
- Refresh flow: refresh token is validated, old refresh token is revoked, and the service rotates a new token pair so long-lived sessions stay safer.
- Protected request flow: JwtAuthenticationFilter extracts the bearer token, validates it, builds the Spring Security Principal, and then protected controllers can rely on Principal for the current user.
- OAuth flow: the OAuth classes convert Google/GitHub identity data into a ConnectSphere user, then return the same JWT-based login model used by local authentication.

File Inventory

The table below lists the important source, configuration, and test files that currently exist in the module. This is meant to help you explain how the module is connected internally when you present it or revise it later.

File	Description and why it matters
pom.xml	Build descriptor defining dependencies, Java version, plugins, and the Spring Boot packaging for auth-service.

File	Description and why it matters
README.md	Human-readable runbook describing what the module does, how to start it, and what endpoints or features it exposes.
src\main\java\com\connectsphere\auth\AuthServiceApplication.java	Spring Boot entry point that starts the module and wires the top-level framework configuration.
src\main\java\com\connectsphere\auth\config\JwtProperties.java	Configuration class that connects framework behavior, routing, security, OpenAPI, scheduling, or websocket support to this module.
src\main\java\com\connectsphere\auth\config\OpenApiConfig.java	Configuration class that connects framework behavior, routing, security, OpenAPI, scheduling, or websocket support to this module.
src\main\java\com\connectsphere\auth\config>PasswordConfig.java	Configuration class that connects framework behavior, routing, security, OpenAPI, scheduling, or websocket support to this module.
src\main\java\com\connectsphere\auth\config\SecurityConfig.java	Configuration class that connects framework behavior, routing, security, OpenAPI, scheduling, or websocket support to this module.
src\main\java\com\connectsphere\auth\controller\AuthResource.java	HTTP or websocket controller that exposes the module's API contract to callers through REST endpoints or STOMP mappings.
src\main\java\com\connectsphere\auth\controller\GlobalExceptionHandler.java	HTTP or websocket controller that exposes the module's API contract to callers through REST endpoints or STOMP mappings.
src\main\java\com\connectsphere\auth\dto\ApiMessageResponse.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\AuthResponse.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\ChangePasswordRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto>LoginRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\LogoutRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.

File	Description and why it matters
src\main\java\com\connectsphere\auth\dto\RefreshTokenRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\RegisterRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\TokenValidationRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\TokenValidationResponse.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\UpdateProfileRequest.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\UserProfileResponse.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\dto\UserSummaryResponse.java	Request/response contract object that keeps transport payloads cleanly separated from internal entities.
src\main\java\com\connectsphere\auth\entity\AuthProvider.java	Persisted domain model representing the data owned by this module.
src\main\java\com\connectsphere\auth\entity\RevokedToken.java	Persisted domain model representing the data owned by this module.
src\main\java\com\connectsphere\auth\entity\Role.java	Persisted domain model representing the data owned by this module.
src\main\java\com\connectsphere\auth\entity\User.java	Persisted domain model representing the data owned by this module.
src\main\java\com\connectsphere\auth\exception\BadRequestException.java	Supporting source file that contributes to the module's implementation, build, or verification story.

File	Description and why it matters
src\main\java\com\connectsphere\auth\exception\NotFoundException.java	Supporting source file that contributes to the module's implementation, build, or verification story.
src\main\java\com\connectsphere\auth\oauth\OAuth2LoginFailureHandler.java	OAuth helper code that maps provider identities into ConnectSphere users and token responses.
src\main\java\com\connectsphere\auth\oauth\OAuth2LoginSuccessHandler.java	OAuth helper code that maps provider identities into ConnectSphere users and token responses.
src\main\java\com\connectsphere\auth\oauth\OAuth2UserProfileService.java	OAuth helper code that maps provider identities into ConnectSphere users and token responses.
src\main\java\com\connectsphere\auth\oauth\OAuthAccountService.java	OAuth helper code that maps provider identities into ConnectSphere users and token responses.
src\main\java\com\connectsphere\auth\oauth\OAuthUserProfile.java	OAuth helper code that maps provider identities into ConnectSphere users and token responses.
src\main\java\com\connectsphere\auth\repository\RevokedTokenRepository.java	Data-access layer used to read and write persisted records through Spring Data JPA.
src\main\java\com\connectsphere\auth\repository\UserRepository.java	Data-access layer used to read and write persisted records through Spring Data JPA.
src\main\java\com\connectsphere\auth\security\AuthenticatedUser.java	Security support code such as JWT parsing, user-details adaptation, or request authentication filters.
src\main\java\com\connectsphere\auth\security\CustomUserDetailsService.java	Security support code such as JWT parsing, user-details adaptation, or request authentication filters.
src\main\java\com\connectsphere\auth\security\JwtAuthenticationFilter.java	Security support code such as JWT parsing, user-details adaptation, or request authentication filters.

File	Description and why it matters
src\main\java\com\connectsphere\auth\security\JwtTokenService.java	Security support code such as JWT parsing, user-details adaptation, or request authentication filters.
src\main\java\com\connectsphere\auth\service\AuthService.java	Business-logic class or contract responsible for orchestrating workflows, validation, integrations, and core rules.
src\main\java\com\connectsphere\auth\service\AuthServiceImpl.java	Business-logic class or contract responsible for orchestrating workflows, validation, integrations, and core rules.
src\main\resources\application.yml	Runtime configuration profile controlling ports, datasource behavior, service-to-service URLs, and environment-specific settings.
src\main\resources\application-mysql.yml	Runtime configuration profile controlling ports, datasource behavior, service-to-service URLs, and environment-specific settings.
src\main\resources\application-oauth.yml	Runtime configuration profile controlling ports, datasource behavior, service-to-service URLs, and environment-specific settings.
src\test\java\com\connectsphere\auth\AuthResourceIntegrationTest.java	Automated test covering this module's runtime behavior and helping verify that important flows remain working.
src\test\java\com\connectsphere\auth\JsonTestHelper.java	Automated test covering this module's runtime behavior and helping verify that important flows remain working.
src\test\java\com\connectsphere\auth\OAuthAccountServiceTest.java	Automated test covering this module's runtime behavior and helping verify that important flows remain working.

Assessment Notes

Current file inventory count: **46**

This guide focuses on how the module works end to end: what it owns, which requests it handles, what supporting files make it run, and how it connects to the rest of ConnectSphere.